

# ARSALAN AMIR ALI

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## EDUCATION

### York University — Lassonde School of Engineering

Sep 2024 — Apr 2029 (Expected)

BSc Computer Science (Specialized Honours) · Specializing in AI · CGPA: 4.62/9.0 · EECS GPA: 5.33/9.0

Awards: International Entrance Award for Academic Excellence · Lassonde Entrance Scholarship

### Beaconhouse School System South

Sep 2022 — Jul 2024

IB Diploma Programme, Computer Science · 27 IB Diploma Points

Awards: Habib University 11th Annual Meritorious Award · 75% Scholarship (Year 2) · 20% Scholarship (Year 1)

## TECHNICAL SKILLS

**Languages** Python · Java · C · JavaScript · SQL

**Backend** FastAPI · Node.js · Express · REST APIs · PostgreSQL · pgvector · Supabase · Redis

**Frontend** Next.js · React

**AI** LLM APIs (Groq, Anthropic Claude, OpenAI) · RAG Pipelines · LangGraph · MCP · Vector Search (Voyage) · Prompt Engineering · PyTorch · Model Fine-Tuning (LoRA/QLoRA) · Triton GPU Kernels

**DevOps** Docker · Railway · Git · GitHub · Linux · CI/CD

**Concepts** Data Structures & Algorithms · OOP · System Design · Software Architecture · API Design

## EXPERIENCE

### AI Software Engineer | Self-Employed — FinSight AI

May 2026 — Jul 2026

- Built and shipped a personal finance intelligence agent independently to production (v1.5.1) — RAG knowledge base over transaction history, LLM-powered spending analysis and personalized recommendations via conversational interface
- Designed stateful agent workflows with LangGraph, Groq (Llama 3.1) chat integration, Voyage embeddings, MCP-connected external tools (currency conversion, market data, budget categories), and pgvector semantic retrieval
- Owned every layer independently — FastAPI backend, PostgreSQL + pgvector, Next.js frontend, Docker containerization; shipped to production on Vercel + Render + Supabase

### Co-Founder & AI Software Engineer | Self-Employed — RoomFinderAI

May 2025 — Present

- Built and deployed AI-powered rental market analysis and negotiation platform on Railway — 60+ REST endpoints, 27 SQL migrations, real-time chat via Supabase Realtime, PWA with push notifications
- Engineered LLM negotiation engine with 8-phase conversation flow, landlord personality modeling, and learning loop trained on 1,247 negotiations achieving 82% win rate on top-performing template
- Architected full auth stack (email, Google OAuth, Apple Sign-In), Stripe payments, Azure identity verification, and 15+ external API integrations per listing including RentCast, FEMA, and BLS

### AI Software Engineer | Self-Employed — KernelForge

Jul 2026 — Present

- Fine-tuning an LLM to generate and optimize Triton GPU kernels from PyTorch operations, scoped to LLM-inference-serving patterns — quantized matmul, attention variants, KV-cache ops, normalization fusion
- Designed and GPU-validated an automated verification harness — correctness checking against PyTorch eager execution, speedup benchmarking, and sandboxed timeout isolation for untrusted generated code
- Building a verification-gated dataset and LoRA fine-tuning pipeline, positioned as a data-efficient, explainable alternative to existing broad-coverage systems (KernelLLM, Kevin-32B)

## PROJECTS

**FinSight AI** — *Python · FastAPI · PostgreSQL · pgvector · Next.js · LangGraph · Groq · Voyage · MCP · Docker*  
[github.com/Arsalan-05/FinSight\\_AI](https://github.com/Arsalan-05/FinSight_AI)

- Live in production (v1.5.1) — RAG pipeline over personal financial data with chunking, embedding, and pgvector retrieval grounding LLM responses in real transaction history for accurate, context-aware spending insights
- Stateful agent with LangGraph — persistent memory, tool use, and MCP-connected external tools; full authentication, financial data ingestion pipeline, and real-time conversational interface

**RoomFinderAI** — *Node.js · Supabase · PostgreSQL · OpenAI · Docker · Railway* [github.com/Arsalan-05/RoomFinderAI](https://github.com/Arsalan-05/RoomFinderAI)

- True cost calculator fusing 15+ external APIs (RentCast, Google Distance Matrix, GreatSchools, FEMA, BLS) — computes real monthly cost including commute, utilities, and insurance per listing
- Property photo AI via Llama 3.2 11B Vision on Cloudflare Workers AI — extracts pricing signals, grades units A+ to C-, and generates suggested rent from feature premiums and flaw penalties

**KernelForge** — *Python · PyTorch · Triton · LoRA/QLoRA (Unsloth) · Kaggle GPU* [github.com/Arsalan-05/KernelForge](https://github.com/Arsalan-05/KernelForge)

- Fine-tuning an LLM to generate and optimize Triton GPU kernels from PyTorch operations, scoped to LLM-inference-serving patterns — quantized matmul, attention variants, KV-cache ops, normalization fusion
- Built and GPU-validated an automated verification harness — correctness checking and speedup benchmarking against PyTorch eager execution, with sandboxed timeout isolation
- In progress: verification-gated dataset curation and LoRA fine-tuning; positioned as a data-efficient, explainable alternative to existing broad-coverage systems (KernelLLM, Kevin-32B)

## EXTRACURRICULAR & INVOLVEMENT

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**York University:** CS Hub · AWS Cloud Club · Data Science Club · AI & Data Science Club · YU Blueprint

**Volunteer:** Event Coordinator – Sports Gala · Technical Crew – Drama Festival · AV Crew – Hackathon 3.0 · Food Stall Coordinator – Flood Relief Carnival